Differential pulse code modulation (DPCM)

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DPCM is a digital communication technique that converts an analog signal into digital form.

→ PCM, which sends the full sample values, DPCM sends only the difference between the current sample & the predicted value based on the previous sample. This helps reduce the amount of data to be transmitted.

Example:- If sample values are :- 10, 12, 15, 17.

PCM would send all values : 10, 12, 15, 17

DPCM would send : 10 (first value) then +2, +3, +2 (differences from previous values)

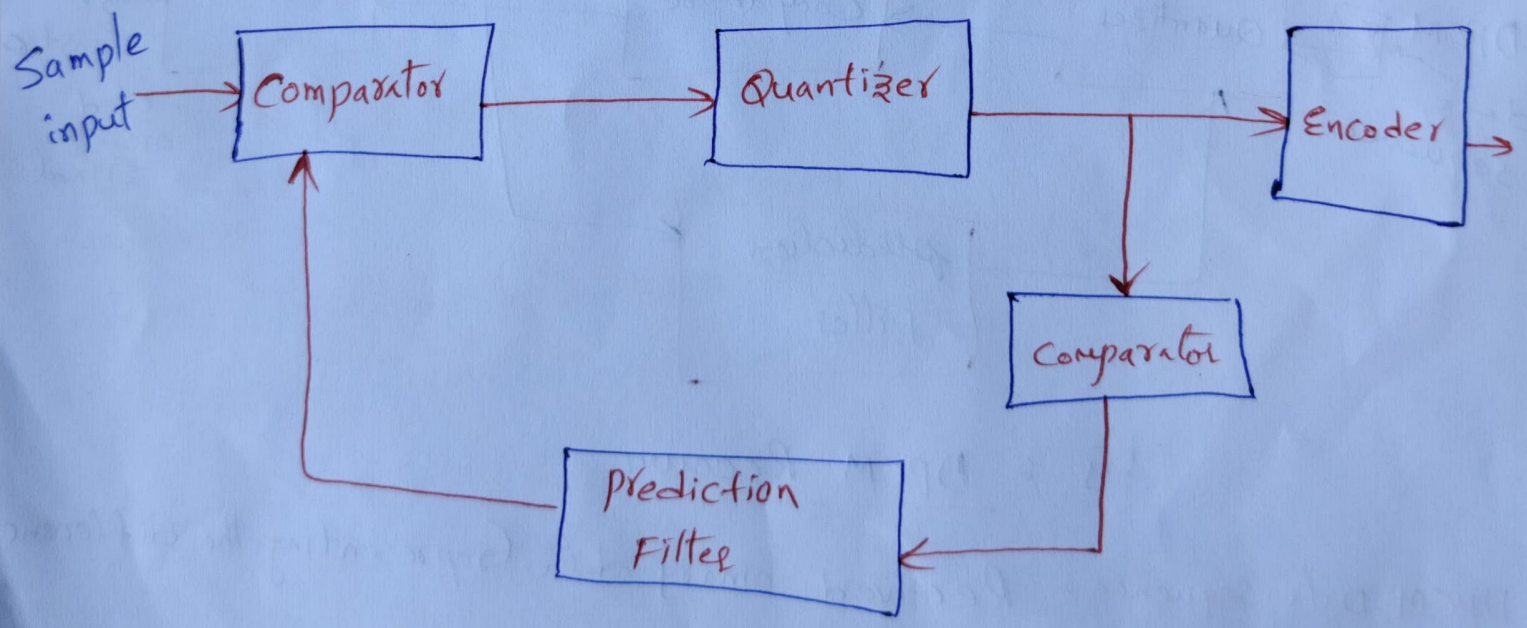

fig : DPCM Transmitter

Sample input :- This is the original analog signal sampled at regular time intervals.

Comparator :- Compares the current sample with the predicted value (from the prediction filter) to find the difference

Quantizer :- Rounds off the difference to the nearest level to reduce the number of bits needed.

Encoder :- Converts the quantized difference into binary form for transmission.

output is the DPCM Data sequence.

prediction filter :- predicts the current sample based on the previous sample (or past samples).

DPCM Receiver :-

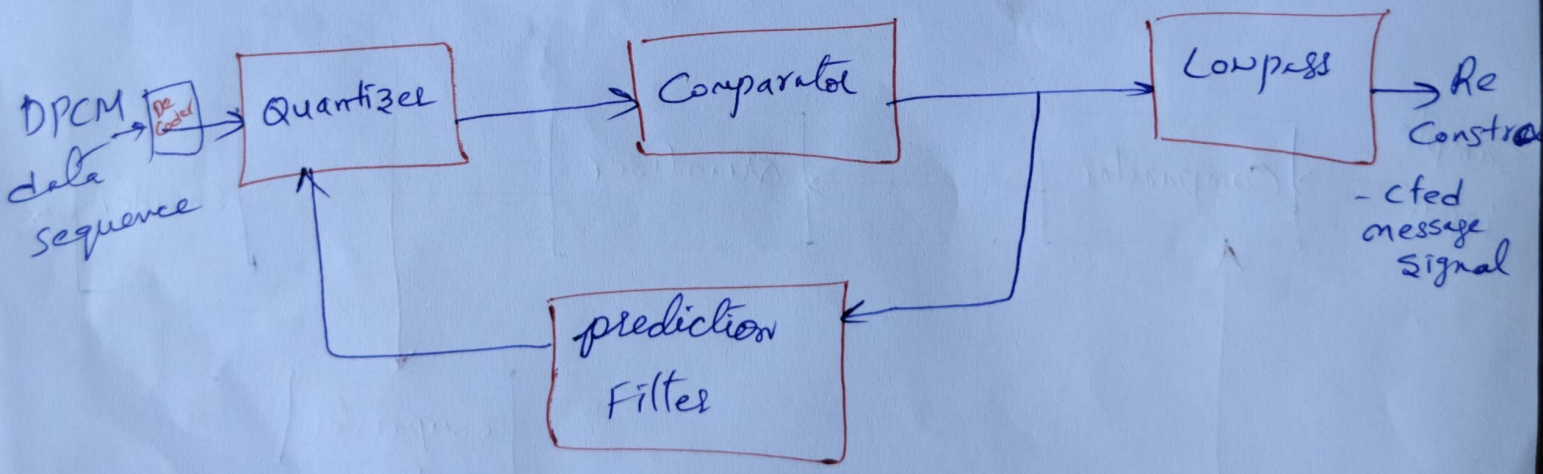

fig :- DPCM Receiver

DPCM Data Sequence :- Received binary data (representing the difference values)

Decoder (assumed before Quantizer in receiver) :- Converts the binary data back to quantized difference values.

Quantizer :- Ensures signal levels are within expected range (may be part of decoder logic)

Comparator (Adder) :- Adds the predicted value (from prediction Filter) to the received difference to reconstruct the actual sample.

Prediction Filter :- Same as in the transmitter - uses previous reconstructed samples to predict the next.

Compass Filter :- Removes high-frequency noise from the reconstructed signal to get smooth message output.

Advantages of DPCM :-

→ Requires less bandwidth than PCM.
→ Reduces the size of the transmitted data.
→ Easier to implement compared to advanced compression methods.

Disadvantages of DPCM :-

→ Errors can occur if the prediction is not accurate.
→ Not suitable for signals that change suddenly.
→ Noise or errors in one step can affect future steps (error propagation).

Applications of DPCM :-

→ Audio and Video compression.
→ Voice communication over the internet (VoIP)
→ streaming over low-bandwidth network.